



Department of Computer Science and Engineering(ICB)

EXPERIMENT 3

Aim: Mongo DB CRUD Command.

Dataset Link: <https://www.kaggle.com/datasets/rtatman/iris-dataset-json-version>

Objective:

To experiment with MongoDB CRUD (Create, Read, Update, Delete) operations and demonstrate how these operations are executed in MongoDB. MongoDB is a NoSQL database that stores data in flexible, JSON-like documents. This write-up aims to provide hands-on examples of how CRUD operations are performed in MongoDB.

1. Environment Setup

Requirements:

- MongoDB instance (either locally installed or via a cloud service like MongoDB Atlas)

MongoDB client (like MongoDB Compass, shell, or any programming language's MongoDB driver)

CRUD Operations in MongoDB

A. Create Operation (Insert Data)

MongoDB provides two methods to insert data: `insertOne()` and `insertMany()`.

- **Insert One Document:**

```
async function createOne() {  
  const user = { name: "John Doe", age: 30, email: "john@example.com" };  
  const result = await collection.insertOne(user);  
  console.log(`A document was inserted with the _id: ${result.insertedId}`);  
}
```

Insert

Multiple Documents:



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```
async function createMany() {  
  const users = [  
    { name: "Alice", age: 25, email: "alice@example.com" },  
    { name: "Bob", age: 28, email: "bob@example.com" }  
  ];  
  const result = await collection.insertMany(users);  
  console.log(`${result.insertedCount} documents were inserted`);  
}
```

B. Read Operation (Query Data)

MongoDB allows querying data using `find()`, which can retrieve a single document or multiple documents.

- **Find One Document:**

```
async function readOne() {  
  const user = await collection.findOne({ name: "John Doe" });  
  console.log(user);  
}
```

Find Multiple Documents:

```
async function readMany() {  
  const users = await collection.find({ age: { $gt: 20 } }).toArray();  
  console.log(users);  
}
```

C. Update Operation (Modify Data)

MongoDB provides methods like `updateOne()`, `updateMany()`, and `replaceOne()` for updating documents.

D. Delete Operation (Remove Data)

MongoDB allows you to delete data using `deleteOne()` and `deleteMany()` methods.

Resources:

<https://www.mongodb.com/docs/compass/master/documents/insert/>

<https://www.mongodb.com/docs/compass/master/documents/mo>



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Screenshots:

Create:

Insert Document

To collection test.test_db

VIEW `{ }`

```
1  _id: ObjectId('67a9b7b0228a64e91d57a9ef')
2  sepalLength : "4.5"
3  sepalWidth  : "3.4"
4  petalLength : "1.2"
5  petalWidth  : "0.3"
   species    : "setosa"
```

ObjectId
String
String
String
String
String

Cancel Insert

Read:



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[+ ADD DATA](#) [EXPORT DATA](#) [UPDATE](#) [DELETE](#)

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a957'),
  "sepalLength": 5.1,
  "sepalWidth": 3.5,
  "petalLength": 1.4,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a958'),
  "sepalLength": 4.9,
  "sepalWidth": 3,
  "petalLength": 1.4,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a959'),
  "sepalLength": 4.7,
  "sepalWidth": 3.2,
  "petalLength": 1.3,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a95a'),
  "sepalLength": 4.6,
  "sepalWidth": 3.1,
  "petalLength": 1.5,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a95b'),
  "sepalLength": 5,
  "sepalWidth": 3.6,
  "petalLength": 1.4,
  "petalWidth": 0.2
}
```

Edit:

```
1  _id: ObjectId('67a9b51d228a64e91d57a957')
2  sepalLength: 5.1
3  sepalWidth: 3.5
4  petalLength: 1.4
5  petalWidth: 0.2
6  species: 'abcd'
```

Document modified.

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a957'),
  "sepalLength": 5.1,
  "sepalWidth": 3.5,
  "petalLength": 1.4,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

Document updated.

Remove:

```
{
  "_id": ObjectId('67a9b51d228a64e91d57a957'),
  "sepalLength": 5.1,
  "sepalWidth": 3.5,
  "petalLength": 1.4,
  "petalWidth": 0.2,
  "species": "setosa"
}
```

Document flagged for deletion.





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×

⚠

Delete 150 documents

test.test_db

Filter ⓘ

{}

⌄

Export

Preview (sample of 5 documents)

_id: ObjectId('67a9b51d228a64e91d57a958')

sepalLength: 4.9

sepalWidth: 3

petalLength: 1.4

petalWidth: 0.2

species: "setosa"

_id: ObjectId('67a9b51d228a64e91d57a959')

sepalLength: 4.7

sepalWidth: 3.2

petalLength: 1.3

petalWidth: 0.2

species: "setosa"

_id: ObjectId('67a9b51d228a64e91d57a95a')

Cancel

Delete 150 documents

Find:



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Filter {sepalLength: 3.1}

+ ADD DATA

EXPORT DATA

UPDATE

DELETE

```
_id: ObjectId('67a9b51d228a64e91d57a958')
sepalLength: 3.1
sepalWidth: 3
petalLength: 1.4
petalWidth: 0.2
species: "setosa"
```

```
_id: ObjectId('67a9b51d228a64e91d57a959')
sepalLength: 3.1
sepalWidth: 3.2
petalLength: 1.3
petalWidth: 0.2
species: "setosa"
```

```
_id: ObjectId('67a9b51d228a64e91d57a95a')
sepalLength: 3.1
sepalWidth: 3.1
petalLength: 1.5
petalWidth: 0.2
species: "setosa"
```

```
_id: ObjectId('67a9b51d228a64e91d57a95b')
sepalLength: 3.1
sepalWidth: 3.6
petalLength: 1.4
petalWidth: 0.2
species: "setosa"
```

```
_id: ObjectId('67a9b51d228a64e91d57a95c')
sepalLength: 3.1
sepalWidth: 3.9
petalLength: 1.7
: 0.4
```

Conclusion: Thus we successfully performed CRUD operations in mongodb using compass.